

Value Added (VA)

A points-equivalent valuation of every box-score contribution, stated in standard mathematical notation.

Version 2 (playoff-tracker) — supersedes the v1 definition shipped in `nba-projects`; divergences itemized in §8.

Authoritative implementations: `app/scoring.js` (client) and `scripts/R/scrape_common.R` (bake), which agree to the decimal.

Value Added answers one question in one unit: **how many points did this player produce, above or below what the league would have produced in the same workload?**

1 NOTATION

1.1 Player observation

A stat line — a game, a playoff run, or a season — is the vector

$$x = (MP, G, PTS, AST, STL, BLK, TOV, DRB, ORB, FGM, FGA, 3PM, 3PA, FTM, FTA)$$

with two-point shooting derived rather than sourced:

$$2PM = FGM - 3PM \qquad 2PA = FGA - 3PA$$

$VA(x) \equiv 0$ when $MP \leq 0$.

1.2 League baselines

For each season y , a baseline vector $\lambda(y)$ is computed once from that season's *player* season-totals table and stored in `app/data/league-averages.json`. Two families of quantity, computed two different ways on purpose (§3).

Per-minute rate baselines — minutes-weighted medians (§2):

$$\mu_{PTS}, \mu_{AST}, \mu_{STL}, \mu_{BLK}, \mu_{TOV}, \mu_{DRB}, \mu_{ORB}$$

Conversion constants — league aggregate ratios over all players in season y :

$$p_3 = \frac{\Sigma 3PM}{\Sigma 3PA} \quad p_2 = \frac{\Sigma 2PM}{\Sigma 2PA} \quad p_F = \frac{\Sigma FTM}{\Sigma FTA} \quad p_G = \frac{\Sigma FGM}{\Sigma FGA}$$

$$\Pi = \Sigma FGA - \Sigma ORB + \Sigma TOV + 0.475 \Sigma FTA$$

possessions — Hollinger estimator

$$\pi = \frac{\Sigma PTS}{\Pi} \quad \kappa = \frac{\Sigma PTS}{\Sigma FGM}$$

points per possession · points per made field goal

$$\rho_D = \frac{\Sigma DRB}{\Sigma TRB} \quad \rho_O = \frac{\Sigma ORB}{\Sigma TRB} = 1 - \rho_D \quad v = \frac{5\Pi}{\Sigma MP}$$

rebound shares · possessions per on-court minute (= pace / 48)

$$\Lambda = \frac{5\Sigma TRB}{\Sigma MP}$$

rebound opportunities per on-court minute, at one end

Λ is used only by the rebound credit (§4.3), v only by the defensive extension (§6). Because ΣMP counts *player* minutes — five per team-minute — the factor of 5 converts to possessions per **team**-minute.

2 THE MINUTES-WEIGHTED MEDIAN

For N players in season y with stat totals X_i and minutes $m_i > 0$, let $r_i = X_i/m_i$ and let $(\cdot)$ denote the ordering $r_{(1)} \leq r_{(2)} \leq \dots \leq r_{(N)}$. Then

$$\mu_X = r_{(k^*)} \qquad k^* = \min \left\{ k : \sum_{j \leq k} m_{(j)} \geq \frac{1}{2} \sum_j m_j \right\}$$

In words: **the rate of the median league minute**. Half of all NBA minutes are played above μ_X , half below.

Why not the two obvious alternatives. The minutes-weighted *mean* (the league aggregate rate) is pulled upward by a handful of high-usage stars, so “average” ends up describing nobody. The unweighted *per-player median* lets a 40-minute call-up outvote a 2,800-minute starter. Weighting the median by minutes fixes both: the baseline is the typical *minute of basketball*, which is exactly the thing VA charges a player for consuming.

3 WHY BASELINES ARE COMPUTED TWO WAYS

The per-minute baselines define “**typical**” — the counterfactual a player is measured against — so they must be robust to skew, hence the weighted median.

The conversion constants ($\pi, \kappa, \rho_D, \rho_O, p_3, p_2, p_F, p_G$) do not define “typical”; they **translate an event into points**. A possession is worth the league’s actual points-per-possession, not the median player’s. These stay aggregate ratios.

4 THE MASTER EQUATION

Every one of the ten categories has the same shape:

$$VA(x, \lambda) = \sum_{c \in C} w_c \ (r_c - \lambda_c) n_c$$

(player rate – league rate) × opportunity × price in points

where C is the ten categories and each term supplies its own rate r_c , baseline λ_c , opportunity n_c , and price w_c :

CATEGORY C	RATE R_C	BASELINE λ_C	OPPORTUNITY N_C	PRICE w_C
Points (volume)	$\frac{PTS}{MP}$	μ_{PTS}	MP	1
3-Pointers	$\frac{3PM}{3PA}$	p_3	3PA	3

CATEGORY C	RATE R_C	BASELINE λ_C	OPPORTUNITY N_C	PRICE w_C
2-Pointers	$\frac{2PM}{2PA}$	p_2	2PA	2
Free Throws	$\frac{FTM}{FTA}$	p_F	FTA	1
Assists	$\frac{AST}{MP}$	μ_{AST}	MP	$\kappa(1 - p_G)$
Steals	$\frac{STL}{MP}$	μ_{STL}	MP	π
Blocks	$\frac{BLK}{MP}$	μ_{BLK}	MP	$\pi \rho_D$
Turnovers	$\frac{TOV}{MP}$	μ_{TOV}	MP	$-\pi$
D Rebounds	$\frac{DRB}{MP}$	μ_{DRB}	MP	$\gamma \pi \rho_O$
O Rebounds	$\frac{ORB}{MP}$	μ_{ORB}	MP	$\gamma \pi \rho_D$

where the rebound credit γ is **per-player**, not a constant — it is the one price that depends on the player’s own rate as well as the league’s. See §4.3.

Attempt-denominators are guarded: $r_c := 0$ when $n_c = 0$, so an unattempted category contributes exactly zero rather than NaN.

4.1 Expanded form — the ten terms

SCORING	
Volume	$(\frac{PTS}{MP} - \mu_{PTS}) \times MP$
3-point shooting	$3 \times (\frac{3PM}{3PA} - p_3) \times 3PA$
2-point shooting	$2 \times (\frac{2PM}{2PA} - p_2) \times 2PA$
Free throws	$(\frac{FTM}{FTA} - p_F) \times FTA$

PLAYMAKING & BALL SECURITY	
Assists	$(\frac{AST}{MP} - \mu_{AST}) \times MP \times \kappa \times (1 - p_G)$
Turnovers	$-(\frac{TOV}{MP} - \mu_{TOV}) \times MP \times \pi$
DEFENSIVE EVENTS	
Steals	$(\frac{STL}{MP} - \mu_{STL}) \times MP \times \pi$
Blocks	$(\frac{BLK}{MP} - \mu_{BLK}) \times MP \times \pi \times \rho_D$
REBOUNDING	
Defensive rebounds	$\gamma \times (\frac{DRB}{MP} - \mu_{DRB}) \times MP \times \pi \times \rho_O$
Offensive rebounds	$\gamma \times (\frac{ORB}{MP} - \mu_{ORB}) \times MP \times \pi \times \rho_D$

4.2 Reading the weights

- **Assist.** An assist is worth the field goal it created, κ , times the share of those baskets that would *not* have happened without the pass, $(1 - p_G)$. Deliberately opinionated: it refuses to credit the passer for the league-baseline conversion the shooter would have managed anyway.
- **Steal vs. block.** A steal ends the possession outright, so it earns a full π . A block only ends it if the defense secures the carom — about ρ_D of the time — so it earns $\pi\rho_D$. The same asymmetry is reused verbatim in the defensive extension (§6.3).
- **Rebound cross-weighting.** A rebound is credited a possession discounted by the probability the team *would not* have gotten it anyway. Defensive boards are the common outcome, so they scale by the smaller ρ_O ; offensive boards are rarer and scale by the larger ρ_D . $\gamma = 5/4$ is not a fitted constant — it is the “one of five claimants” correction derived in §4.3.
- **Volume and efficiency are separate terms, not a net.** A high-usage scorer at league efficiency has done something real (absorbed difficulty); a low-usage scorer at elite efficiency has also done something real. Netting them erases both. VA pays for each.

4.3 The rebound credit γ

γ answers a question the other nine terms never have to ask: **a rebound is the one box-score event that is guaranteed to be allocated and rivalrous.** Every miss produces exactly one rebound, and exactly one of ten players on the floor gets it. So “would this have happened anyway?” has a real answer.

The contest. Under a Luce / Bradley–Terry model, give each of the five defenders a claim weight d and each of the five offensive players a claim weight o . Then $P(\text{defense secures}) = 5d/(5d+5o) = \rho_D$, and the team's securing **odds** are $\Omega = \rho_D/\rho_O = d/o$. Remove one of the five defenders and $\Omega' = 4d/5o = (4/5)\Omega$, so the odds of *losing* the possession rise by exactly 5/4 — independent of ρ , of era, and of which team is rebounding.

But VA multiplies a probability, not an odds, and the two differ. Working in probability space, and generalizing off the “exactly one fifth” assumption to a player who claims a fraction q of the boards available at his end:

$$\gamma = \frac{1}{1 - q} \quad q = \frac{\text{REB}/\text{MP}}{\Lambda}$$

$$\Lambda = \text{laREBoppPerM}, 0.9071 \text{ in } 2025\text{-}26$$

Every miss is rebounded at one end or the other, so a player on the floor sees Λ chances per minute at his own end. The construction is self-checking: $d/(d+o)$ recomputed from these same sums returns ρ_D exactly.

ρ cancels out of the correction. Substituting $q = sp$ (where s is the player's share of his *team's* boards) into $1/(1 - sp)$ gives the same expression with no ρ in it. ρ still prices the board — $\pi\rho_O$ for a defensive rebound, $\pi\rho_D$ for an offensive one — but it plays no part in the credit. One formula covers both ends.

It needs no team data. Λ is a league quantity, so q comes straight off the player's own line. This matters: every other VA term reads only the player's stat line and league baselines, which is what makes a 1987 season comparable to a 2026 one. A team-relative γ would break that — measured on 2025-26, players inside a narrow DRB/min band saw 5–7% γ swings from teammates alone, so identical production would have scored differently on different rosters. It doesn't here.

Special cases. The league-average rebounder has $q = 0.2\rho$, giving

$$\gamma = \frac{1}{1 - 0.2\rho} = \frac{5}{5 - \rho} \quad \Rightarrow \quad \gamma_{\text{DRB}} \approx 1.174 \quad \gamma_{\text{ORB}} \approx 1.053$$

— the “one of five” constant, and the fallback for any season baked before Λ existed. The **odds-space** $5/4 = 1.25$ that VA shipped through July 2026 is the $\rho \rightarrow 1$ limit of that expression: correct as an odds ratio, but used as though it were a probability, which overstated defensive credit ~6% and offensive credit ~19%. At the top of the distribution the per-player form runs the other way — a player covering a third of his end reaches $\gamma \approx 1.5$.

Estimation. q is unshrunk by design, matching every other VA category (none of which is shrunk). A season is 1,400–2,300 opportunities, putting binomial error on γ under 1.5%; a single game is ~32 and swings hard, exactly as one night's shooting does in the efficiency terms. q is clamped to $[0, 0.5]$ so $\gamma \leq 2$ stays finite on 1-minute garbage-time lines — the only rows where the clamp ever binds.

4.4 What γ assumes

Three assumptions remain, each a place the model could still be wrong.

1. **Independence of irrelevant alternatives.** Removing a claimant redistributes his share proportionally across the other nine. Real rebounding is closer to matched pairs — the opposing big absorbs most of a removed center's share, not the floor uniformly.
2. **A league-uniform opportunity rate.** Λ is a league constant, but the boards available at a given player's end depend on his team's pace and his opponents' shooting. Using the team's own rate instead moves γ by a median of 0.74% (p90 2.11%, $r = 0.985$) — less than the binomial noise already in q , and it would cost the context-freedom above. Deliberately traded away.
3. **The marginal board is an average board.** q counts an uncontested carom the same as a won contest, and most defensive rebounds are uncontested. The boards a strong rebounder adds are drawn from the contested tail, where recovery risk is far above p_O — which argues every γ here is *understated* for players who actually box out. This is the one objection no version of the model addresses; closing it needs contested-rebound data the box score does not carry.

Two open items. μ_{DRB} (minutes-weighted median) sits ~11% below the league aggregate rate because rebounding is right-skewed and positionally concentrated, so the median baseline already tilts this term toward bigs and γ widens the same spread. Separately, the league earns roughly **1.8× more positive VA from offensive rebounds than defensive ones** — two similarly-sized populations of over-performers paid at $p_D/p_O \approx 2.85$ different rates. That is a baseline question, not a γ question, and it remains open.

4.5 Rate forms

$$VAG = \frac{VA}{G} \qquad VA/36 = 36 \frac{VA}{MP}$$

5 EXACT DECOMPOSITION

The category vector is the summand of the master equation:

$$VA_c(x, \lambda) = w_c (r_c - \lambda_c) n_c \qquad \sum_{c \in C} VA_c = VA$$

This identity is a **product requirement**, not a happy accident: every UI surface — leaderboard total, compare-panel header, stacked category bars, percentile context, comp-shape vectors — is computed from the same ten

terms, so the parts always reconcile to the headline to the decimal.

The per-game category vector used downstream (§7) is

$$v(x) = \left(\frac{VA_c(x, \lambda)}{G} \right)_{c \in C} \in \mathbb{R}^{10}$$

6 VA⁺ — THE DEFENSIVE EXTENSION

Box-score defense is two events wide. VA⁺ adds a fifth category built from *points actually allowed*.

6.1 Bayesian rating

Each player-season carries two defensive ratings: D_{est} , the box-score estimate, and D_{pbp} , points allowed per 100 possessions while on the floor (play-by-play, 2000-01+). With defended possessions $\Pi_p = \text{MP} \cdot v$ and prior strength Π_0 ,

$$\theta = \frac{\Pi_p}{\Pi_p + \Pi_0} \quad \hat{D} = \theta D_{\text{pbp}} + (1 - \theta) D_{\text{est}} \quad \Pi_0 = \begin{cases} 1500 & \text{regular season} \\ 500 & \text{playoffs} \end{cases}$$

The estimate is an *informed prior worth Π_0 possessions of evidence*; real possessions overtake it as they accrue. A full-time starter lands ~75% on play-by-play; a 300-minute season stays mostly prior — which is what stops a small-sample on-court fluke from topping the leaderboard. The lighter playoff prior reflects both the smaller samples available and the noisier estimate behind them. Seasons with no play-by-play source are simply all-prior ($\theta = 0$).

The team baseline $\hat{T}$ blends with the **same** θ , so the individual term below subtracts like from like.

6.2 Net rating

With league line $L = 100\pi$ and team edge $E = L - \hat{T}$:

$$\text{net} = (\hat{T} - \hat{D}) + w E$$

IND: edge over his own team + TM⁺: share of the team's edge vs. league

6.3 The earned share

Splitting team defense five ways equally credits a passenger like an anchor. The share is instead *earned*, with stocks priced exactly as VA prices them — a block is worth ρ_D of a steal:

$$s = \frac{\text{STL} + \rho_D \text{BLK}}{\text{MP}} \quad w_e = \text{clamp} \left(0.2 \frac{s}{s_{\text{team}}}, 0.05, 1 \right)$$

$$w = \begin{cases} w_e & E \geq 0 \text{ — credit is **earned**} \\ \text{clamp} (0.4 - w_e, 0.05, 1) & E < 0 \text{ — blame **shrinks** with activity} \end{cases}$$

The bad-defense branch is the earned share mirrored around the 1-in-5 split: contesting shields you from a collective failure, passivity draws more of it. Both branches conserve the team pot up to the clamps and are continuous at $E = 0$. Multi-team rows fall back to the plain vs-league form ($w = 1$).

6.4 Defensive value

$$\text{dVA} = \frac{\text{net}}{100} \cdot v \cdot \text{MP} \quad \text{VA}^+ = \text{VA} + \text{dVA}$$

VA^+ is undefined — and hidden — when neither rating source exists.

7 DERIVED INSTRUMENTS

7.1 Shot-zone value (1996-97+)

For zone $z \in \{ 0\text{-}3 \text{ ft}, 3\text{-}10 \text{ ft}, 10\text{-}16 \text{ ft}, 16 \text{ ft-}3\text{PT} \}$ with league zone accuracy p_z — the 2-point term of §4, parameterized by distance:

$$\text{VA}_z = 2 \left(\frac{M_z}{A_z} - p_z \right) A_z$$

Held deliberately **outside** C : shot-location data begins in 1996-97, and folding it into the core vector would punch holes in every earlier season's decomposition and career totals.

7.2 Closest comps

For player-seasons a, b with vectors $v(a), v(b)$ (§5):

$$\cos(a,b) = \frac{\langle v(a), v(b) \rangle}{\|v(a)\| \|v(b)\|} \quad \text{mag}(a,b) = \frac{\min(\|v(a)\|, \|v(b)\|)}{\max(\|v(a)\|, \|v(b)\|)}$$

$$S(a,b) = \cos(a,b) \times \text{mag}(a,b)$$

archetype match $\times$ level match

Admissible only under $|MPG_a - MPG_b| \leq 7$ (same minutes role), $G \geq 8$, and $\cos \geq 0.3$ — a clearly different archetype is never a comp.

The **Shooting** lens applies the identical $\cos \times \text{mag}$ form to the 6-dimensional shooting profile (4 zones + 3-Pointers + Free Throws), additionally gated on shot diet: writing $\delta(x) = 3PA/FGA$ for the share of a player's field-goal attempts taken from three, a comp is admissible only when $|\delta(a) - \delta(b)| \leq 0.15$. Without that gate, two players with identical (zero) 3-point *impact* read as shooting twins even when one lives at the rim and the other is a high-volume league-average bomber.

7.3 Draft scoring — the app's origin

Sixteen playoff teams are drafted eight apiece. For a won series in round r with base $B = (1, 2, 4, 8)$ for rounds 1–4, and seeds σ :

$$P_{\text{series}} = B_r + \max(0, \sigma_{\text{winner}} - \sigma_{\text{loser}})$$

the seed differential pays only on an upset

and a live series projects linearly in games won:

$$\hat{P}_{\text{series}} = P_{\text{series}} \cdot \frac{\text{games won}}{4}$$

with real results and user “what-if” speculation accumulated in separate ledgers so the two are never confused.

7.4 Championship leverage and Legacy

§4 deliberately prices a possession the same in October and June. That is what makes a season comparable to a season. It is also what makes career VA the wrong instrument for an all-time ranking, which has to answer two questions VA declines to: whether a game mattered, and whether a career was tall or long.

Legacy answers both **outside** C , exactly as §7.1 does. It reweights an existing per-game VA decomposition and never re-scores anything, so §9.1 holds unchanged on every existing surface and no bake is affected.

Implementations: `app/lib/leverage.js` and `app/lib/legacy.js`.

7.4.1 Game leverage

For a series in state a – b (the score **before** the game) under a best-of- n format, let $W(a,b)$ be the probability of winning the series with every game an independent coin:

$$W(a,b) = \frac{1}{2} W(a+1,b) + \frac{1}{2} W(a,b+1)$$

$$W(k,\cdot) = 1, W(\cdot,k) = 0, k = \lfloor n/2 \rfloor + 1$$

The game’s **series swing** and its **championship leverage** are

$$\sigma(a,b) = W(a+1,b) - W(a,b+1), \quad \text{cLI} = \sigma(a,b) \cdot 2^{-\rho}$$

$$\rho = \text{rounds still to be won after this one}$$

In a best-of-7, σ is 0.3125 at 0-0, 0.375 at 1-1 and 2-1, 0.25 at 2-0, 0.5 at 2-2 and 3-2, 0.125 at 3-0, and exactly 1 at 3-3 — a Game 7 swings the whole series by construction, which is the identity the rest rests on.

Neutrality ($p = \frac{1}{2}$) is deliberate, and is the same commitment §4 makes: a player is not paid more for having been on a favourite, and the quantity needs no seed, venue or roster data, so it is portable to any era.

The ordering this produces is not the obvious one:

GAME	Σ	P	CLI	WEIGHT AT $A = \frac{1}{2}$
R1 G1	0.3125	3	0.0391	1.000
R1 G5 at 2-2	0.5	3	0.0625	1.265

GAME	Σ	P	CLI	WEIGHT AT A = ½
R2 G1	0.3125	2	0.0781	1.414
R1 G7	1.0	3	0.1250	1.789
R2 G7	1.0	2	0.2500	2.530
Finals G1	0.3125	0	0.3125	2.828
Finals G7	1.0	0	1.0000	5.060

A Round-2 opener outranks a tied Round-1 Game 5, but only by 1.25×; a Round-1 Game 7 outranks both. Nothing here is asserted — it falls out of $\sigma \cdot 2^{-p}$.

7.4.2 The unit, and the regular season

Weights are expressed against **that season's own opening playoff game**, for a bracket of depth d :

$$cLI_{\text{ref}} = \sigma(0,0) \cdot 2^{-(d-1)}, \quad w = \left(\frac{cLI}{cLI_{\text{ref}}} \right)^\alpha$$

Anchoring per season rather than globally is what keeps eras honest: a 1960 title required two series won, not four, so a global anchor would pay the eight-team league roughly 4× for clearing a shallower field. For 1980-2026 every bracket is four rounds deep and $cLI_{\text{ref}} = 0.0390625$ throughout, so the correction is a no-op on the current corpus and matters only under backfill.

The regular season is priced in the same currency rather than by a separate dial. Across a full season a team travels from undetermined to holding a berth, and a berth is worth 2^{-d} of a title:

$$cLI_{\text{RS}} = \frac{2^{-d}}{G_{\text{season}}} = \frac{0.0625}{82} = 0.000762$$

$$\Rightarrow w_{\text{RS}} = 0.13969 \text{ at } \alpha = \frac{1}{2}$$

About 1/51 of a playoff opener raw, about 1/7 after compression. G is tabulated, not measured: $\max(g)$ over a season's totals overshoots, because a traded player's TOT row sums his games across teams (1999-00 and 2003-04 both report 85).

α is **the one free parameter in the construction, and it is a taste parameter, not a derivation**. $\alpha = 1$ is raw championship leverage, under which a Finals Game 7 is worth 25.6 openers and the regular season all but

disappears; $\alpha = 0$ is flat and reproduces context-free VA exactly. The default $\frac{1}{2}$ says importance counts at diminishing returns.

$$\text{LVA}(\text{season}) = \sum_{g \in \text{playoffs}} w(g) \cdot \text{VA}(g) + w_{\text{RS}} \cdot \text{VA}_{\text{RS}}$$

7.4.3 Peak and longevity

Order a player's seasons by LVA descending and fold with geometric decay:

$$\text{Legacy} = \sum_{k \geq 1} D^{k-1} \cdot \text{LVA}_{(k)}$$

$D = 1$ is a plain career sum (pure longevity); $D \rightarrow 0$ keeps only the best season (pure peak). Every term is positive when the season was, so extra seasons only ever add — but they add geometrically less, so a long tail of replacement-level years is neither rewarded nor punished much.

D is pinned rather than chosen, by stating what share of a career the peak should carry. “The best K of N seasons carry half the weight” is

$$\frac{1 - D^K}{1 - D^N} = \frac{1}{2}$$

$$\Rightarrow D = 0.938068 \text{ at } (K, N) = (7, 20)$$

hence the default $D = 0.94$.

The second axis is the **rate**: LVA per weighted game across the best K seasons. Weighted, not raw — if leverage prices the total it must price the rate, or the two halves of the metric disagree about what a game is worth. The two axes are reported side by side rather than blended, because any blend needs both normalized against the pool leader, which would make a career number move when an unrelated rookie debuts.

7.4.4 What this does to the Jordan/LeBron question

Over their best seven seasons Jordan leads on raw per-game VA, 23.19 to 21.75. He does not lead once the games are priced by what was at stake: the two cross at $\alpha \approx 0.27$, and at the default $\alpha = \frac{1}{2}$ LeBron leads 21.84 to 21.26. The crossing sits in the same place on best-8 (0.279) and best-10 (0.257), so it is not an artifact of K .

On the volume axis there is **no** crossing at all: LeBron leads at every D from 0.5 to 1, because his best leveraged seasons are also better, and the margin widens with D . The dials move the board substantially further down —

Harden falls from 3rd to 23rd as α runs 0 to 1, Magic climbs 13th to 4th, Duncan 16th to 7th — which is the metric separating profiles, and is worth reading as the result rather than tuning away.

7.4.5 A known inexactness

Legacy sums the **per-game** VA decomposition. That is not identical to the baked season-total VA: γ (§4.3) is non-linear in REB/MP, so evaluating it once on season totals differs from evaluating it per game and summing — a mean gap of 1.8% on meaningful seasons, systematically upward for rebounding bigs, up to +11.9 (1980-81 Moses Malone). Legacy takes the per-game aggregation, on the grounds that γ is defined on a stat line and a game *is* a stat line. So $\alpha = 0$ reproduces a career’s *summed per-game* VA exactly, and its summed *season-total* VA only to about 1.8%.

7.4.6 Coverage, and what a backfill has to solve

Legacy currently spans **1980-81 to 2025-26**, the seasons with player data on disk. A career already running in 1980-81 is measured short and is flagged as such rather than ranked as complete; Magic Johnson and Larry Bird sit in the top 15 missing their rookie year, and Kareem Abdul-Jabbar appears with 848 of his games.

Extending back is a data problem, not a formula problem. The leverage construction already generalizes: bracket depth is read per season, so a two-round 1950s postseason scores its Finals Game 7 at the same cLI = 1, and the per-season anchor of §7.4.2 makes its opening game worth exactly the same as a modern one. Byes need no special case. What does not generalize is the **category set**:

ERA	STATE	WHAT IT NEEDS
1979-80	complete; the 3-point line arrives	the existing bake, run
1973-74 – 1978-79	complete box score, no 3-point line ($\lambda_3 = 0$ against 3PA = 0, so the term is exactly zero)	the existing bake; verify TOV provenance, which BR carries per player only from 1977-78
1970-71 – 1972-73	baselines are zero-filled and unusable — $\mu_{\text{DRB}} = \mu_{\text{ORB}} = \rho_{\text{D}} = 0$, because rebounds were not split until 1973-74	a reduced-category variant, and rebuilt baselines
1949-50 – 1969-70	no baselines at all; no rebound split, no steals/blocks/turnovers, possessions need imputing	v1’s VA_3 and its OLS possession estimate, both dropped in v2

The third row is a live §9.5 violation sitting in `League-averages.json` already: those entries exist and are wrong, and a zero baseline is not a harmless zero — at $\mu_{\text{DRB}} = 0$ a player is credited his entire rebounding rate as surplus, and at $\rho_{\text{D}} = 0$ the block term multiplies to nothing. Nothing reads them today because no player data predates 1980-81, so `baselineCoverage()` in `app/scoring.js` refuses any season whose baseline cannot

price all ten categories, and Legacy skips it loudly rather than scoring it. That is the tripwire the backfill will meet first.

The comparability caveat is the real cost, and it should not be buried: a season scored on seven of ten categories is **not the same currency** as one scored on ten, and \$9.5 forbids making up the difference. Any backfilled era that loses categories needs to carry how many it measured, and be shown as its own class rather than mixed into one column.

8 V1 → V2 DIVERGENCES

nba-projects (R Shiny, 2024) established the framework; playoff-tracker (2026) revised it. Numbers from the two apps are **not** interchangeable.

	V1 — NBA-PROJECTS	V2 — PLAYOFF-TRACKER
Per-minute baseline	unweighted median across players	minutes-weighted median (\$2)
Possessions	$2PA + 3PA + TOV + \frac{FTA}{2.1}$ pre-1973 imputed by OLS on μ_{PTS}, μ_{TRB}	$FGA - ORB + TOV + 0.475 FTA$ Hollinger
Rebound credit	$\gamma = 1$	per-player $\gamma = 1/(1 - q)$ (\$4.3); a flat 5/4 through July 2026
Coverage	1949-50+, via three nested variants — VA_1 full; VA_2 drops TOV for 1974-77; VA_3 drops TOV/STL/BLK and estimates rebound splits for 1952-73 — selected by <code>coalesce()</code>	1979-80+, single variant, complete box score
Playoffs	separate tab	first-class scope, scored against that season's regular-season baselines
Defense	box-score events only	VA^+ (\$6)
Shot location	Four Factors visuals	zone VA + Shooting comps (\$7.1–7.2)
Non-NBA	—	NCAA D-I on college-derived baselines
All-time ranking	career VA, summed	Legacy : championship-leverage weighted, peak/longevity dialled (\$7.4)

1. $\sum_c VA_c = VA$, to the decimal, on every surface.
2. Baselines are **season-local**: a 1987 season is scored against 1987, never against today. Playoff runs use their own season's regular-season baselines, so October and June are the same currency.
3. Non-NBA populations get **their own** λ ; the formula is portable, the baselines are not.
4. A season is scored by one $\lambda(y)$ from one source table; a plausibility gate ($0.33 < \mu_{PTS} < 0.52$) refuses to write a baseline derived from a mis-parsed page.
5. Any category the source does not carry is **absent**, never zero-filled — a missing measurement must not read as below-average performance.

Companion document: **portfolio-summary.md** — the executive summary of the framework, its theses, and the systems built on it.

Self-contained: this page loads no external fonts, scripts, or stylesheets. All notation is rendered in HTML and CSS.